

Intelligent Remote Healthcare and Assisted Living Ecosystem

A PROJECT REPORT

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ABSTRACT

Monitoring the health and safety of elderly people, patients, and individuals who need regular assistance is becoming increasingly important. Family members and caregivers may not always be available to monitor them continuously. Because of this, sudden changes in health, falls, gas leakage, or other emergency situations may sometimes go unnoticed. To address this problem, the Intelligent Remote Healthcare and Assisted Living Ecosystem is developed as an integrated system for health monitoring, safety monitoring, and emergency assistance. The system uses an ESP32 microcontroller to collect information from different sensors. The MAX30102 sensor measures heart rate and blood oxygen level (SpO₂), while the MLX90614 sensor monitors body temperature. The MPU6050 sensor is used to identify unusual movements and possible falls. Along with health monitoring, the system also checks the surrounding environment. The DHT22 sensor measures room temperature and humidity, the MQ-2 sensor detects gas or smoke, and the PIR sensor detects movement in the surroundings. The collected sensor readings are displayed on an OLED screen and are also sent to the Blynk IoT platform, allowing the user's condition to be monitored remotely. If an abnormal condition or possible emergency is detected, the system provides an alert through a buzzer and LED indicators. An SOS button is also provided, allowing the user to manually request assistance during an emergency. To maintain a record of the monitored information, a Python-based data logging system is included. The Python program retrieves the sensor data from the Blynk IoT Cloud and automatically stores the readings in a CSV file. This stored data can be used for reviewing previous readings, analyzing changes over time, and maintaining health and environmental records. Overall, the proposed system combines health monitoring, environmental safety, emergency alerts, remote access, and automatic data storage in one platform. It provides a practical and affordable approach that can help family members and caregivers stay informed about the user's condition and take appropriate action when assistance is needed.

KEYWORDS:

Remote Healthcare, Assisted Living, ESP32, Health Monitoring, Heart Rate Monitoring, SpO₂ Monitoring, Body Temperature Monitoring, Fall Detection, Environmental Monitoring, Emergency Alert, Blynk IoT, Python Data Logging.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

1.SYMBOLS:

Symbol	Meaning
SpO ₂	Oxygen level
HR	Heart Rate
T	Temperature
°C	Degree Celsius

V	Voltage
C	Current
BPM	Beats Per Minute

2.ABBREVIATIONS:

Abbreviation	Full Form
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
BPM	Beats Per Minute
SpO ₂	Blood Oxygen Saturation
HR	Heart Rate
HRM	Heart Rate Monitoring
MLX90614	Infrared Non-Contact Temperature Sensor
MAX30102	Pulse Oximeter and Heart Rate Sensor
ESP32	Espressif 32-bit Microcontroller

3.NOMENCLATURE:

Keyword	Simple Meaning
IoT	Connecting devices to the internet to send and receive data.
ESP32	A small microcontroller that controls the whole system and connects to Wi-Fi.
Remote Healthcare	Monitoring a person's health from a different location.

Health Monitoring	Checking important health conditions regularly.
Heart Rate	The number of heartbeats in one minute.
SpO ₂	The amount of oxygen present in the blood.
MAX30102	Sensor used to measure heart rate and SpO ₂ .
MLX90614	Sensor used to measure body temperature without touching the body.
MPU6050	Sensor used to detect movement and possible falls.
Fall Detection	Identifying sudden movement that may indicate a person has fallen.
DHT22	Sensor used to measure room temperature and humidity.
MQ-2	Sensor used to detect gas and smoke.
OLED Display	Small screen used to show sensor readings.

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CHAPTER 1

INTRODUCTION

1.1 Introduction to Remote Healthcare

Healthcare monitoring is important for elderly people, patients, and individuals who need regular assistance. In many situations, family members or caregivers cannot stay with them throughout the day. Because of this, changes in health conditions or unexpected situations may not be noticed immediately. A remote healthcare system can help by continuously collecting important health and environmental information and making it available to caregivers from a distance. The Intelligent Remote Healthcare and Assisted Living Ecosystem is developed to provide continuous monitoring using sensors and IoT technology. An ESP32 microcontroller collects information from different sensors such as the MAX30102, MLX90614, MPU6050, DHT22, MQ-2, and PIR sensor. These sensors are used to monitor heart rate, SpO₂, body temperature, movement, room conditions, gas or smoke, and surrounding activity. The collected information can be displayed locally on an OLED display and remotely monitored through the Blynk IoT platform. The system also provides alerts when an unusual condition is detected. A buzzer and LED are used to indicate abnormal situations, while an SOS button allows the user to manually request assistance. The system is designed as a practical and affordable solution for supporting caregivers and improving the safety of people who require regular monitoring.

1.2 Need for Elderly and Patient Monitoring

Elderly people and patients may require regular observation because their health condition can change unexpectedly. Continuous personal monitoring is not always possible, especially when family members or caregivers are away. Simple changes such as an unusual heart rate, low SpO₂ level, increased temperature, a fall, or unsafe environmental conditions may require immediate attention. A monitoring

system can reduce this problem by observing important parameters continuously. Instead of depending completely on manual checking, sensors can collect information automatically and send it to a remote platform. This allows caregivers to check the person's condition without being physically present all the time. The proposed system also considers the surroundings of the person. Room temperature and humidity, gas or smoke, and movement can provide useful information about the safety of the environment. Combining health monitoring with environmental monitoring makes the system more useful for assisted living applications.

1.3 Problems with Conventional Healthcare Monitoring

Traditional healthcare monitoring often depends on direct observation by family members, nurses, or caregivers. This method can become difficult when the person needs monitoring for a long period. Caregivers may not always be available to check the person's condition continuously.

Another problem is that unexpected events such as falls or gas leakage may happen when nobody is nearby. In such situations, a delay in identifying the problem can make the situation more serious. Manual recording of health readings can also take additional time and may not provide an easy way to review previous readings.

These limitations show the need for a system that can automatically collect sensor readings, provide alerts, allow remote monitoring, and store data for later reference.

1.4 Objectives of the Project

1. To develop a system for continuous monitoring of important health parameters.
2. To measure heart rate and SpO₂ using the MAX30102 sensor.

3. To monitor body temperature using the MLX90614 sensor.
4. To identify unusual movement or possible falls using the MPU6050 sensor.
5. To monitor room temperature and humidity using the DHT22 sensor.
6. To detect gas or smoke using the MQ-2 sensor.
7. To detect surrounding movement using the PIR sensor.
8. To display sensor readings locally using an OLED display.
9. To provide remote monitoring through the Blynk IoT platform.
10. To generate alerts using a buzzer and LED when abnormal conditions are detected.
11. To provide an SOS button for manually requesting assistance.
12. To store collected sensor data in CSV format using Python for future review and analysis.

1.5 Methodology

The proposed Intelligent Remote Healthcare and Assisted Living Ecosystem follows an integrated IoT-based methodology for continuous health and environmental monitoring. Multiple sensors, including the MAX30102, MLX90614, DHT22, MQ-2, PIR, and MPU6050, are used to collect vital health and surrounding environmental parameters. The ESP32 acts as the central processing unit, receiving and processing the sensor data and comparing critical parameters with predefined threshold values to identify abnormal conditions. The processed information is displayed locally through an OLED display and transmitted through Wi-Fi to the Blynk IoT Cloud, enabling remote monitoring. When an emergency condition such as abnormal vital signs, gas detection, fall detection, or SOS activation is identified, the system provides immediate alerts through the buzzer, LED, and Blynk platform. In addition, a Python-based data logging system retrieves the monitoring data from Blynk Cloud and stores it in CSV format with timestamps, enabling historical analysis and performance

evaluation. This methodology provides a low-cost and integrated approach for remote healthcare monitoring and assisted living applications.

CHAPTER 2

LITERATURE SURVEY

The increasing need for independent living and continuous healthcare support for

older adults has encouraged the development of IoT, artificial intelligence, remote monitoring, and assistive technologies. Existing research shows a clear shift from conventional healthcare, which mainly depends on periodic clinical visits, toward continuous and technology-assisted monitoring within home environments. These developments provide the foundation for the proposed Intelligent Remote Healthcare and Assisted Living Ecosystem.

Baldissera et al. (2024) developed the PRO-@GE collaborative communication and monitoring ecosystem to support elderly care by connecting elderly individuals, families, caregivers, professionals, and care organizations. The platform combines monitoring with collaborative communication and was implemented and validated as an ecosystem rather than as an isolated monitoring device. The study highlights that effective elderly care requires not only sensor-based monitoring but also timely communication and coordinated responses among caregivers. This approach is particularly relevant to the proposed system, which combines sensor monitoring, emergency alerts, and cloud-based remote access.

Aalam, Shah, and Parveen (2025) discussed personalized healthcare services within the concept of Healthcare 5.0, emphasizing the role of intelligent and personalized technologies in assisted living. Their work reflects the transition toward healthcare systems that can use collected information to provide more individualized support rather than relying solely on conventional healthcare delivery. This provides an important conceptual basis for integrating multiple patient parameters into a unified monitoring platform, as proposed in the present work.

Maskeliūnas, Damaševičius, and Segal (2019) reviewed IoT technologies for Ambient Assisted Living (AAL) environments and highlighted the potential of connected sensing devices to improve safety, independence, and quality of life. Their work demonstrates how IoT can connect physical devices and monitoring

systems within assisted-living environments, while also identifying challenges associated with implementation and reliable operation.

Qian et al. (2021) presented a comprehensive review of AI-enabled IoT technologies for elderly assisted living and healthcare monitoring. The study examined different approaches for combining artificial intelligence with IoT-based sensing and discussed their applications, advantages, limitations, and future directions. The authors emphasized the importance of intelligent interpretation of sensor information for supporting elderly people living independently. This supports the proposed architecture in which sensor data is collected through the ESP32 and transmitted to a cloud-based platform for continuous monitoring.

Ahmad et al. (2022) investigated emerging technologies for next-generation remote healthcare and assisted living. Their work highlights the importance of remote healthcare in improving accessibility, sustainability, and the continuity of healthcare services. The study brings together technologies associated with sensing, communication, IoT, and intelligent healthcare systems, showing that remote monitoring can reduce dependence on conventional physical healthcare interactions. These findings support the use of IoT connectivity in the proposed system for transmitting patient and environmental information to remote caregivers.

Abdi, de Witte, and Hawley (2020) conducted a scoping review of emerging technologies with potential applications in elderly care. Their review identified technologies including AI-enabled health applications, wearable devices, portable diagnostics, intelligent homes, voice-based systems, robotics, and extended-reality technologies. Importantly, the study also noted that older people and other stakeholders were not consistently involved in the design of emerging technologies. This indicates that technical capability alone is insufficient; usability and user involvement are also essential for successful assisted-living solutions.

Sapci and Sapci (2019) systematically reviewed assisted-living tools, remote monitoring technologies, AI-based solutions, and robotic systems for ageing societies. Their findings indicate that continuous monitoring and telecare technologies have significant potential for supporting older adults and reducing the risk of undetected health problems. The review also emphasizes the importance of evidence-based development and evaluation of such technologies. This motivates the proposed system's combination of vital-sign monitoring, environmental sensing, fall detection, and emergency notification.

Uddin (2026) presented a data-centric perspective on smart assisted living, focusing on how sensor data can be transformed into meaningful information using machine learning and intelligent home technologies. The work describes smart homes as systems capable of perceiving environmental and behavioural conditions, learning patterns, predicting situations, and supporting adaptive responses. This recent direction strengthens the importance of collecting structured historical data, which is addressed in the proposed system through Python-based data logging and CSV storage.

Timon et al. (2024) investigated the development of independent-living support for older adults using IoT and AI through a co-design approach. Their work reinforces the importance of involving users and stakeholders when developing technology intended for independent living. Rather than treating technology as a standalone solution, the study considers how IoT and AI can be incorporated into systems that genuinely support older adults in their daily lives. This provides a strong human-centred foundation for the proposed healthcare and assisted-living ecosystem.

Research Gap

The reviewed studies demonstrate strong progress in IoT-based monitoring, AI-assisted healthcare, ambient assisted living, remote communication, and intelligent

homes. However, many existing approaches focus primarily on one aspect, such as elderly communication, environmental monitoring, AI-based interpretation, or remote healthcare. There remains a practical need for a low-cost, integrated prototype that combines vital-sign monitoring, environmental sensing, fall detection, emergency alerts, cloud-based remote monitoring, and historical data logging within a single system. The proposed Intelligent Remote Healthcare and Assisted Living Ecosystem addresses this gap by integrating these functions using an ESP32-based sensing unit, Blynk IoT Cloud, local alert mechanisms, and Python-based data logging.

CHAPTER 3

PRINCIPLES OF THE PROPOSED SYSTEM

3.1 Health Parameter Monitoring

Health parameter monitoring is one of the main functions of the proposed system. The system measures important parameters such as heart rate, SpO₂, and body temperature using suitable sensors.

The MAX30102 sensor is used to measure heart rate and SpO₂, while the MLX90614 sensor is used to measure body temperature. The ESP32 collects the readings from these sensors and processes them before displaying the values.

Regular monitoring of these parameters can help caregivers keep track of the person's condition. The system is mainly intended for monitoring and alert purposes and does not replace professional medical equipment or medical diagnosis.

3.2 Heart Rate and SpO₂ Measurement

The MAX30102 sensor is used to measure heart rate and blood oxygen saturation (SpO₂). It works using optical sensing, where light is used to detect changes in blood flow. When the user's finger is placed properly on the sensor, it collects the required signals and sends the data to the ESP32.

The ESP32 processes the received data and obtains the heart rate and SpO₂ values. These values can then be shown on the OLED display and sent to the Blynk platform for remote viewing.

For better readings, the finger should be placed correctly on the sensor and unnecessary movement should be avoided during measurement. This helps the sensor provide more stable readings.

3.3 Body Temperature Measurement

The MLX90614 sensor is used to measure body temperature without direct contact with the body. It works by detecting infrared radiation from the surface of the target and converting it into a temperature value.

The sensor is positioned properly in front of the person to obtain the temperature reading. The measured value is received by the ESP32 and can be displayed on the OLED screen and monitored through the Blynk platform.

This method provides a simple and convenient way to monitor temperature. However, the distance, position of the sensor, and surrounding conditions can affect the measured value.

3.4 Fall Detection Principle

Fall detection is an important part of the proposed system, especially for elderly people and individuals who may require assistance. The MPU6050 sensor is used to monitor movement and changes in body orientation. It contains an accelerometer and gyroscope that provide information about movement.

During normal activities, the movement of a person changes gradually. A fall may cause a sudden change in acceleration or orientation. The ESP32 continuously receives the movement information from the MPU6050 and checks it according to the conditions programmed in the system.

When a possible fall or unusual movement is detected, the system can activate the buzzer and LED to attract attention. This feature is intended to provide an early alert so that the person can receive assistance when required.

CHAPTER 4

SYSTEM DESIGN AND COMPONENT DESCRIPTION

4.1 Overall System Architecture

The overall system consists of three main sections: sensing, processing, and monitoring. The sensing section includes the MAX30102, MLX90614, MPU6050, DHT22, MQ-2, and PIR sensors. These sensors collect different types of information from the user and the surrounding environment.

The ESP32 acts as the main processing unit. It receives the sensor readings, processes them according to the programmed conditions, and sends the required information to the output devices. The OLED display provides local information, while the Blynk IoT platform is used for remote monitoring.

The buzzer and LED are connected to the ESP32 to provide alerts when an abnormal condition is detected. The SOS button gives the user an additional way to request help manually.

Table 4.1 Components List

S. No.	Component	Function
1	ESP32 Development Board	Main controller and Wi-Fi communication
2	MAX30102	Heart rate and SpO ₂ measurement
3	MLX90614	Body temperature measurement
4	MPU6050	Movement and fall monitoring
5	DHT22	Room temperature and humidity measurement
6	MQ-2	Gas and smoke detection

7	PIR Sensor	Surrounding movement detection
8	OLED Display	Local display of sensor readings
9	Buzzer	Audible alert
10	LED	Visual alert
11	SOS Button	Manual assistance request

4.2 Component Description

4.2.1 ESP32 Development Board

The ESP32 is the main controller used in the project. It receives information from all the connected sensors and processes the readings according to the program. It also has built-in Wi-Fi, which makes it suitable for sending data to the Blynk IoT platform. The ESP32 controls the OLED display, buzzer, LED, and SOS button along with the different sensors. Its compact size and built-in wireless communication make it suitable for this type of IoT-based project.



Figure 4.1

4.2.2 MAX30102 Sensor

The MAX30102 is used to measure heart rate and SpO₂. It is a small optical sensor that can detect changes in blood flow using light. The sensor sends the collected information to the

ESP32. The processed heart rate and SpO₂ values are then displayed and made available for remote monitoring.

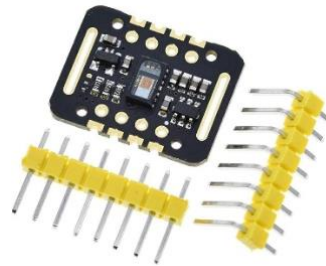


Figure 4.2

4.2.3 MLX90614 Sensor

The MLX90614 is an infrared temperature sensor used to measure body temperature without direct contact. It detects infrared radiation from the target and provides a temperature reading. In this project, the sensor is used to monitor the user's body temperature. The reading is sent to the ESP32 and can be displayed on the OLED and monitored through Blynk.

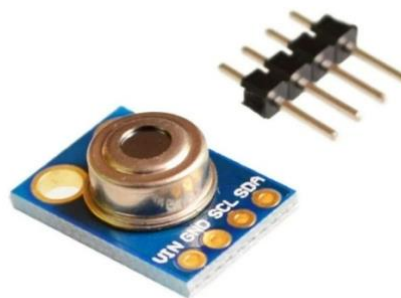


Figure 4.3

4.3.4 MPU6050 Sensor

The MPU6050 is a motion sensor containing an accelerometer and gyroscope. It provides information about acceleration and changes in orientation. The sensor is

mainly used in this project for movement monitoring and possible fall detection. The ESP32 analyzes the movement information and activates the alert system when the programmed fall conditions are detected.



Figure 4.4

4.2.5 DHT22 Sensor

The DHT22 is used to measure the temperature and humidity of the surrounding room. It provides digital output, which makes it convenient to connect with the ESP32. The readings from the DHT22 help in understanding the indoor environmental conditions of the monitored person.



Figure 4.5

4.2.6 MQ-2 Gas Sensor

The MQ-2 sensor is used to detect the presence of gases and smoke in the surrounding environment. It is included in the system to improve environmental

safety. When the gas or smoke level reaches the programmed detection condition, the ESP32 can activate the buzzer and LED to provide a warning.



Figure 4.6

4.2.7 PIR Sensor

The PIR sensor is used to detect movement in the surrounding area. It works by sensing changes in infrared radiation caused by moving objects or people. In the proposed system, the PIR sensor provides information about activity around the monitored person. This can be useful for basic surrounding movement monitoring.



Figure 4.7

4.2.8 OLED Display

The OLED display is used to show the sensor readings locally. Important values such as heart rate, SpO₂, temperature, and environmental readings can be displayed on the screen. The display allows the user or nearby caregiver to check the information directly without depending completely on the mobile application.



Figure 4.8

4.2.9 Buzzer

The buzzer is used as an audible warning device. When the system detects a programmed abnormal condition, the ESP32 activates the buzzer to attract attention. It provides a simple and immediate indication that the monitored person or surrounding environment may require attention.



Figure 4.9

4.2.10 LED

The LED provides a visual indication when an alert condition occurs. It works along with the buzzer to make the warning easier to notice. The LED can also be programmed to indicate specific system conditions depending on the requirements of the project.



Figure 4.10

4.2.11 SOS Button

The SOS button is provided for manual assistance. If the user feels uncomfortable or needs help, the button can be pressed to send the programmed emergency signal. The ESP32 detects the button press and activates the required alert function. This provides the user with a simple way to request assistance without waiting for an automatic sensor-based alert.



Figure 4.11

4.3 System Advantages

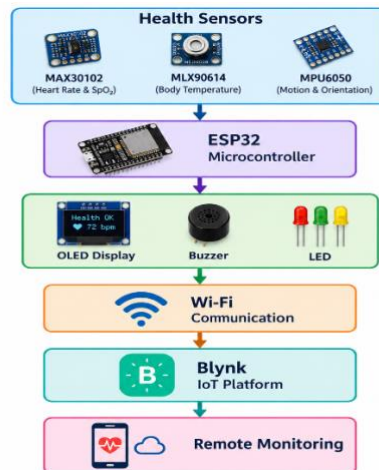
- Continuous monitoring of selected health parameters.
- Monitoring of room temperature and humidity.
- Detection of gas or smoke.
- Movement and possible fall monitoring.
- Remote monitoring through the Blynk platform.
- Local readings through an OLED display.
- Automatic alerts using a buzzer and LED.

CHAPTER 5

CIRCUIT DESIGN AND CONNECTIONS

5.1 Circuit Diagram

The circuit diagram shows how the different sensors and output devices are connected to the ESP32. The health monitoring sensors are connected to measure the required body parameters, while the environmental sensors are used to monitor the surrounding conditions. The MAX30102, MLX90614, and MPU6050 mainly provide health and movement-related information. The DHT22, MQ-2, and PIR sensors are used for environmental and surrounding monitoring. The OLED display is connected to show the sensor readings locally.



The DHT22, MQ-2, and PIR sensors are also connected to the ESP32 to provide environmental information.

5.2 Electrical Connections

The electrical connections are made between the ESP32 and the individual sensors and output devices. Proper power and signal connections are important to prevent incorrect readings and communication problems.

The sensors are supplied with the required operating voltage, and their ground terminals are connected to the common ground of the ESP32. Digital and I²C

communication pins are used according to the requirements of each sensor. The MAX30102, MLX90614, MPU6050, and OLED display use the I²C communication method.

Therefore, their SDA and SCL lines are connected to the corresponding I²C pins of the ESP32. The DHT22 and other digital sensors are connected to suitable GPIO pins. The buzzer, LED, and SOS button are connected to digital GPIO pins. All connections should be checked carefully before powering the circuit.

Table 5.1 Sensor and ESP32 Wiring

Component	ESP32 Connection	Purpose
MAX30102	I ² C SDA/SCL	Heart rate and SpO ₂
MLX90614	I ² C SDA/SCL	Body temperature
MPU6050	I ² C SDA/SCL	Movement and fall detection
OLED Display	I ² C SDA/SCL	Display readings
DHT22	Digital GPIO	Room temperature and humidity
MQ-2	Analog/Digital GPIO	Gas or smoke detection
PIR Sensor	Digital GPIO	Movement detection
Buzzer	Digital GPIO	Audible alert
LED	Digital GPIO	Visual alert
SOS Button	Digital GPIO	Manual assistance

CHAPTER 6

WORKING PRINCIPLE

6.1 Health Monitoring Process

The health monitoring section measures important parameters such as heart rate, SpO₂, and body temperature. The MAX30102 sensor is used to measure heart rate and SpO₂, while the MLX90614 sensor is used to measure body temperature.

When the user places a finger on the MAX30102 sensor, the sensor collects the required optical signals and sends the data to the ESP32. The controller processes the information and obtains the corresponding heart rate and SpO₂ values.

The MLX90614 measures the user's temperature using infrared sensing. The ESP32 receives this reading and processes it along with the other health data.

The measured values are displayed on the OLED screen and transmitted to the Blynk platform. This allows both the user and caregiver to view the monitored health information.

6.2 Fall Detection Process

The fall detection section is designed to identify sudden and unusual body movements. The MPU6050 sensor is used for this purpose because it contains an accelerometer and gyroscope.

The sensor continuously measures changes in acceleration and orientation. During normal activities, these values generally change within a reasonable range. A sudden change in movement can indicate a possible fall or unusual physical activity.

The ESP32 receives the MPU6050 readings and compares them with the conditions defined in the program. When the readings indicate a possible fall, the system activates the buzzer and LED to provide a local warning.

The fall detection function is mainly intended to provide an early alert. It does not replace professional medical or emergency services.

6.3 Environmental Monitoring Process

The system also monitors the environment around the user. This is useful because unsafe environmental conditions can affect the person's comfort and safety.

The DHT22 sensor measures room temperature and humidity. The MQ-2 sensor is used to detect gas or smoke, while the PIR sensor detects movement in the surrounding area.

The ESP32 continuously receives information from these sensors. If the gas or smoke level reaches the programmed detection condition, the system can activate the buzzer and LED. The room temperature, humidity, and movement information can also be monitored through the OLED display and Blynk platform.

This combination of sensors allows the system to monitor both the user's condition and the surrounding environment.

6.4 Alert Generation Process

The alert system is used to provide a quick warning when an abnormal or emergency condition is detected. The ESP32 checks the sensor readings continuously and compares them with the conditions defined in the program.

If a possible fall, gas or smoke detection, or another programmed abnormal condition occurs, the buzzer is activated to produce an audible warning. At the same time, the LED provides a visual indication of the alert condition.

The SOS button provides a manual way for the user to request assistance. When the button is pressed, the ESP32 detects the input and activates the programmed alert function.

The alert mechanism helps nearby people become aware of the situation quickly and provides an additional safety feature for the monitored person.

CHAPTER 7

EXPERIMENTAL SETUP AND IMPLEMENTATION

7.1 Prototype Description

The prototype is developed by connecting all the required sensors and output devices to the ESP32. The system is designed to monitor both the person and the surrounding environment.

- ESP32 Development Board
- MAX30102 Heart Rate and SpO₂ Sensor
- MLX90614 Body Temperature Sensor
- MPU6050 Accelerometer and Gyroscope
- DHT22 Temperature and Humidity Sensor
- MQ-2 Gas and Smoke Sensor
- PIR Motion Sensor
- OLED Display
- Buzzer
- LED
- SOS Push Button

The MAX30102 is used to measure heart rate and blood oxygen level. The MLX90614 is used for non-contact body temperature measurement. The MPU6050 monitors movement and helps identify possible falls. The DHT22 measures room temperature and humidity, while the MQ-2 detects gas or smoke. The PIR sensor is used to detect movement in the surrounding area.

All the sensor readings are processed by the ESP32. The collected information is then displayed and transmitted to the Blynk IoT platform for remote monitoring.

7.2 Hardware Setup

The hardware setup is assembled according to the designed circuit connections. The ESP32 is placed at the centre of the prototype because it acts as the main processing and communication unit.

The MAX30102 and MLX90614 sensors are positioned so that they can be used easily for collecting health-related measurements. The MPU6050 is fixed properly so that changes in movement and orientation can be detected more effectively.

The DHT22, MQ-2, and PIR sensors are arranged to monitor the surrounding environment. The OLED display is placed where the readings can be viewed easily. The buzzer and LED are connected to the ESP32 to provide audible and visual alerts. The SOS button is also connected to a digital input of the ESP32.

Proper wiring and stable power connections are maintained during the setup. Before testing the complete system, each sensor is checked individually to make sure that it is working correctly.

7.3 Sensor Integration

After completing the hardware connections, the sensors are integrated with the ESP32. Each sensor is programmed according to its required communication method and function.

The MAX30102, MLX90614, MPU6050, and OLED display use the I²C communication interface. The DHT22, MQ-2, and PIR sensors are connected to suitable GPIO pins of the ESP32.

The ESP32 continuously reads the sensor values and processes them according to the programmed conditions. The health-related values include heart rate, SpO₂, and body temperature. The environmental values include room temperature, humidity, gas or smoke detection, and surrounding movement.

The sensors are first tested separately and then operated together. This helps in identifying wiring or communication problems before performing the complete system test.

7.4 Blynk and Data Logging Setup

The Blynk IoT platform is used to provide remote access to the sensor information. The ESP32 sends the collected data to Blynk through its Wi-Fi connection. The required virtual pins and dashboard elements are configured to display the sensor readings. The Blynk dashboard can show values such as:

- Heart rate
- SpO₂ level
- Body temperature
- Room temperature
- Humidity
- Gas or smoke status
- Motion status
- Fall alert status

The dashboard allows the readings to be observed remotely without depending only on the OLED display. A Python-based data logging system is also included in the project. The Python program retrieves the available sensor information from the Blynk IoT Cloud and stores it in a CSV file. This makes it possible to maintain a record of previous readings. The stored data can later be reviewed to observe changes in the user's health and environmental conditions over a period of time.

7.5 Testing Procedure

The prototype is tested step by step to verify the operation of each part of the system. Initially, the ESP32 and individual sensors are checked to confirm that they are communicating correctly.

The MAX30102 is tested by placing a finger on the sensor and observing the heart rate and SpO₂ readings. The MLX90614 is tested by placing a person or suitable object within its sensing range and checking the temperature reading.

The MPU6050 is tested by changing its orientation and movement to check whether movement conditions are detected. A controlled movement condition is also used to verify the possible fall-alert response.

The DHT22 is tested under different room conditions to observe changes in temperature and humidity. The MQ-2 is checked for gas or smoke detection, while the PIR sensor is tested by detecting movement in front of the sensor.

The buzzer and LED are tested to ensure that they respond when the programmed alert conditions occur. The SOS button is also pressed during testing to verify that a manual assistance request is detected by the ESP32.

Finally, the complete system is operated together to check whether sensor readings, alerts, Blynk monitoring, and data logging work properly as a combined system.

7.6 Experimental Observation

During the experimental testing, the prototype is observed under different operating conditions. The main purpose of the observation is to verify whether the sensors provide readings, whether the ESP32 processes the data correctly, and whether the monitoring and alert functions respond as expected.

The health sensors provide readings that can be viewed on the OLED display and Blynk dashboard. The environmental sensors provide information about the surrounding conditions. When a programmed abnormal condition is detected, the buzzer and LED provide an alert.

The MPU6050 is used to identify sudden or unusual movement that may indicate a possible fall. The SOS button provides an additional manual method for requesting assistance.

The Blynk platform helps in viewing the collected information remotely, while the Python data logging system stores the readings in CSV format. This provides a useful record for later review and comparison.

Overall, the experimental setup demonstrates the basic operation of the proposed system by combining health monitoring, environmental monitoring, movement detection, alert generation, remote access, and data storage. The system is intended as a monitoring and early-alert solution and should not be considered a replacement for professional medical equipment or medical diagnosis.

CHAPTER 8

COST ESTIMATION

8.1 Cost Estimation Methodology

The cost estimation for the proposed system is carried out by identifying all the major hardware components required for its development. Each component is considered based on its function, quantity, and approximate market price.

The major components include the ESP32 microcontroller, MAX30102, MLX90614, MPU6050, DHT22, MQ-2, PIR sensor, OLED display, buzzer, LEDs, and SOS button. Additional items such as connecting wires, breadboards, resistors, and power supply components may also be required during the assembly and testing of the prototype.

The estimated price of each component is recorded separately, and the individual costs are added to obtain the approximate total hardware cost. The prices may vary depending on the supplier, component quality, availability, and purchase quantity. Therefore, the calculated amount should be considered an approximate cost for developing the prototype rather than a fixed commercial price.

Table 8.1 Component-Wise Cost Breakdown

S. No.	Component	Quantity	Approximate Cost (₹)
1	ESP32 Development Board	1	450
2	MAX30102 Sensor	1	150
3	MLX90614 Sensor	1	300
4	MPU6050 Sensor	1	120
5	DHT22 Sensor	1	150

6	MQ-2 Gas/Smoke Sensor	1	100
7	PIR Motion Sensor	1	80
8	OLED Display	1	150
9	Buzzer	1	20
10	LED Indicators	2	20
11	SOS Push Button	1	20
12	Breadboard and Jumper Wires	1 set	200
13	Power Supply/USB Cable	1	150
Estimated Total			₹1,910

8.3 Economic Feasibility

The proposed system is economically feasible because it uses a combination of low-cost and easily available electronic components. The ESP32 provides sufficient processing capability and built-in wireless connectivity, which reduces the need for additional communication modules.

Using multiple sensors with a single ESP32 also helps reduce the overall hardware requirement. The system can monitor health parameters, detect possible falls, monitor environmental conditions, generate emergency alerts, and send data to the cloud using the same central controller.

Another advantage is the use of Blynk IoT and Python-based data logging, which provides remote monitoring and data storage without requiring expensive

dedicated monitoring equipment. The stored data can also be used for reviewing previous measurements and observing changes over time.

Considering the number of functions provided and the relatively low prototype cost, the proposed system offers a practical and affordable solution for basic remote healthcare and assisted living applications. With further development, the system could be improved using a compact enclosure, rechargeable power supply, improved sensor calibration, and additional communication or monitoring features while still maintaining its focus on affordability.

CHAPTER 9

RESULTS AND ANALYSIS

9.1 Heart Rate Results

The MAX30102 sensor was used to measure the user's heart rate. During testing, the sensor was placed properly on the fingertip to obtain a stable reading. The measured heart rate was displayed on the OLED screen and was also sent to the Blynk IoT platform.

For a normal resting condition, the measured heart rate was observed to be within a reasonable range. Small variations in the readings were noticed depending on finger placement, movement, and the time required for the sensor to obtain a stable signal.

Table 9.1 BPM Condition

Test No.	Observed Heart Rate (BPM)	Condition
1	72	Resting
2	75	Resting
3	78	Resting
4	74	Resting
5	76	Resting

9.2 SpO₂ Results

The MAX30102 sensor was also used to measure the blood oxygen saturation level (SpO₂). The sensor uses optical measurements to estimate the oxygen saturation in

the blood. During testing, the finger was placed over the sensor and the readings were observed after allowing sufficient time for the value to stabilize.

Table 9.2 SpO₂ Condition

Test No.	Observed SpO ₂ (%)	Condition
1	98	Resting
2	97	Resting
3	98	Resting
4	97	Resting
5	98	Resting

The observed readings were generally stable during the test. Minor changes were noticed due to finger movement and sensor positioning. The results confirm that the MAX30102 can be integrated into the proposed system for continuous SpO₂ monitoring.

9.3 Body Temperature Results

The MLX90614 infrared temperature sensor was used to measure body temperature without direct physical contact. The sensor was positioned at an appropriate distance from the body, and the temperature readings were observed through the OLED display and Blynk dashboard.

Table 9.3 Temperature Condition

Test No.	Observed Temperature (°C)	Condition
1	36.5	Normal

2	36.7	Normal
3	36.6	Normal
4	36.8	Normal
5	36.6	Normal

The readings showed only small variations during repeated measurements. The accuracy of the temperature measurement can be affected by the distance between the sensor and the target, surrounding temperature, and the position of the sensor.

9.4 Fall Detection Results

The MPU6050 sensor was used to identify sudden movements that may indicate a possible fall. The sensor measures acceleration and changes in movement along different axes. During testing, different movements were performed to check whether the system could distinguish normal movement from sudden movement.

Table 9.4 Movement Condition

Test No.	Movement	Detection Result
1	Normal sitting	No fall detected
2	Normal walking	No fall detected
3	Sudden movement	Alert condition detected
4	Simulated fall	Fall detected
5	Standing and sitting	No fall detected

When a sudden movement similar to a fall was detected, the system activated the alert mechanism. The buzzer and LED indicators were used to notify the user or nearby caregiver.

The results show that the MPU6050 can be used for basic fall detection. However, certain sudden movements may produce readings similar to a fall. Therefore, the fall detection algorithm can be further improved by using additional parameters and more advanced motion analysis.

9.5 Environmental Monitoring Results

The proposed system also monitors the surrounding environment to improve the safety of the user. The DHT22, MQ-2, and PIR sensors were tested individually under different conditions.

The DHT22 sensor was used to measure room temperature and humidity. The readings were displayed on the OLED screen and transmitted to Blynk.

The MQ-2 sensor was tested by exposing it to a small amount of smoke or gas under controlled conditions. When the sensor detected an increased gas or smoke level, the system generated an alert.

The PIR sensor was tested by detecting movement within its sensing range. When movement was detected, the corresponding status was updated and the system responded according to the programmed conditions.

9.6 Blynk Monitoring Results

The Blynk IoT platform was used to provide remote access to the sensor readings. The ESP32 transmitted the collected data through a Wi-Fi connection to the Blynk IoT Cloud.

The Blynk dashboard was configured to display important parameters such as:

- Heart rate

- SpO₂ level
- Body temperature
- Room temperature
- Humidity
- Gas or smoke status
- Motion status
- Fall alert
- SOS status

During testing, the sensor values displayed on the Blynk dashboard were successfully updated when new readings were received from the ESP32.

The remote monitoring feature makes it possible for a caregiver or family member to observe the user's condition without being physically present near the monitoring device. The effectiveness of this feature depends on the availability of a stable Wi-Fi connection.

9.7 Python Data Logging Results

A Python-based data logging system was implemented to maintain a record of the sensor readings. The Python program retrieves the required information from the Blynk IoT Cloud and stores it in a CSV file.

The recorded data includes parameters such as heart rate, SpO₂, body temperature, room temperature, humidity, and other relevant system statuses

A typical data record is stored in a format similar to the following:

Table 9.5 Data Record

Time	Heart Rate	SpO₂	Body Temp.	Room Temp.	Humidity
10:00:01	80	98	36.5	29.3	61
10:01:01	85	97	36.6	29.4	62
10:02:01	83	98	36.7	29.4	62

9.8 Alert System Results

The alert system was tested under different emergency and abnormal conditions. The buzzer and LED indicators were used to provide a local warning when an alert condition was detected.

The following conditions were tested:

Table 9.6 Test Condition

Test Condition	Alert Generated	Result
Normal health readings	No	Successful
Possible fall	Yes	Successful
Gas/smoke detected	Yes	Successful
Emergency movement	Yes	Successful
SOS button pressed	Yes	Successful
Normal environmental condition	No	Successful

When an emergency condition was detected, the buzzer was activated and the LED indicator provided a visual warning. The SOS button also allowed the user to manually trigger an emergency alert.

The results indicate that the alert mechanism responds effectively to the programmed conditions and provides an additional safety feature for the user.

9.9 Overall System Performance

The complete system was tested by operating the health monitoring, environmental monitoring, remote monitoring, alert, and data logging functions together. The ESP32 successfully collected data from the connected sensors and processed the readings according to the programmed conditions.

The OLED display provided local information, while the Blynk IoT platform provided remote access to the monitored parameters. The alert devices responded when abnormal conditions were simulated, and the Python program successfully recorded the received data in a CSV file.

Overall, the prototype performed the main functions for which it was designed. The integration of multiple sensors into a single system makes it possible to monitor both the user's health and the surrounding environment. The combination of local alerts, cloud-based monitoring, and data storage also improves the usefulness of the system for assisted living applications.

9.10 Limitations

1. **Sensor accuracy:** The readings from low-cost sensors may not be as accurate as those obtained from medical-grade instruments.
2. **Sensor positioning:** Sensors such as the MAX30102 and MLX90614 require proper positioning to obtain more stable measurements.

3. **Fall detection:** The MPU6050-based method may sometimes identify sudden normal movements as possible falls. More advanced algorithms could improve the accuracy.
4. **Wi-Fi dependency:** Remote monitoring through Blynk requires a stable internet or Wi-Fi connection. If the connection is unavailable, real-time cloud monitoring may be interrupted.
5. **Power supply:** The system requires a reliable power source for continuous operation. A backup battery could be added for better reliability.
6. **Environmental factors:** Temperature, humidity, sensor placement, and surrounding conditions can affect the readings of some sensors.
7. **Prototype size:** Since the project is developed as a prototype using separate modules, the complete system may be larger than a final wearable or commercial device.
8. **Medical usage:** The proposed system is intended for monitoring and assistance purposes. It should not be treated as a substitute for professional medical diagnosis or certified medical equipment.

Despite these limitations, the developed prototype demonstrates the feasibility of combining health monitoring, environmental safety, emergency alerts, IoT-based remote monitoring, and automatic data logging into a single affordable system.

CHAPTER 10

ADVANTAGES AND APPLICATIONS

10.1 Advantages

1. **Continuous Monitoring:**

The system can continuously monitor important parameters such as heart rate, SpO₂, and body temperature. This reduces the need for frequent manual checking.

2. **Remote Access:**

Sensor readings can be viewed remotely through the Blynk IoT platform. This allows family members or caregivers to keep track of the user's condition even when they are not nearby.

3. **Fall Detection:**

The MPU6050 sensor helps identify sudden movements that may indicate a fall. This feature can be particularly useful for elderly people who may require immediate assistance after a fall.

4. **Environmental Safety:**

The system monitors room temperature, humidity, gas or smoke, and movement. This provides additional protection against certain environmental hazards.

5. **Emergency Alert:**

The buzzer and LED indicators provide immediate local alerts when an abnormal condition is detected. The SOS button also allows the user to manually request assistance.

6. **Automatic Data Storage:**

The Python-based data logging system stores sensor readings in a CSV file. This makes it easier to review previous measurements and observe changes over time.

7. Low-Cost Implementation:

The system uses commonly available and relatively affordable electronic components. This makes the prototype suitable for low-cost healthcare and assisted living applications.

8. Multiple Functions in One System:

Health monitoring, environmental monitoring, fall detection, emergency alerts, IoT connectivity, and data logging are combined into a single platform.

9. Easy Monitoring:

The OLED display provides immediate access to important readings near the user, while the Blynk dashboard provides a convenient method for remote monitoring.

10. Expandable Design:

Since the system is based on an ESP32 microcontroller, additional sensors and features can be integrated in the future according to the requirements of the application.

10.2 Applications

1. Elderly Care:

The system can assist in monitoring elderly people who live alone or require regular supervision. Fall detection and emergency alerts can provide additional safety.

2. Patient Monitoring:

It can be used for basic monitoring of patients at home, allowing caregivers or family members to observe selected health parameters remotely.

3. Assisted Living Facilities:

The system can be adapted for use in assisted living environments where several individuals may require regular monitoring.

4. Home Healthcare:

The proposed system can support home-based healthcare by providing basic health measurements and remote access to sensor data.

5. Safety Monitoring:

Gas or smoke detection, movement detection, and emergency alerts make the system useful for improving safety in residential environments.

6. Remote Caregiving:

Family members or caregivers can use the Blynk dashboard to check the monitored information without being physically present at the user's location.

7. Healthcare Monitoring Research:

The prototype can also be used as a development platform for experimenting with IoT-based healthcare monitoring, sensor integration, and data analysis.

8. Smart Home Applications:

The environmental monitoring and emergency alert features can be integrated with other smart home systems to improve safety and automation.

10.3 Future Scope

The proposed system can be further improved by adding more advanced features and improving the accuracy and reliability of the existing functions.

Future versions could include GPS tracking to provide the user's location during an emergency, particularly for elderly people who may move outside the home. A GSM or other cellular communication module could also be added to send emergency notifications when Wi-Fi is unavailable.

The fall detection system can be improved using more advanced algorithms or machine learning techniques to distinguish actual falls from normal movements.

Similarly, sensor calibration and signal processing can be improved to obtain more stable health measurements.

A rechargeable battery and power management system could be incorporated to make the device portable and suitable for longer periods of operation. The overall hardware could also be made smaller by designing a compact PCB instead of using separate modules.

Another possible improvement is the development of a mobile application with customized notifications and historical graphs. The collected data could also be analyzed using data analytics or machine learning techniques to identify long-term changes in health parameters.

With these improvements, the proposed system could be developed into a more compact, reliable, and user-friendly solution for remote healthcare and assisted living applications.

11. APPENDICES

11.1 Prototype Images

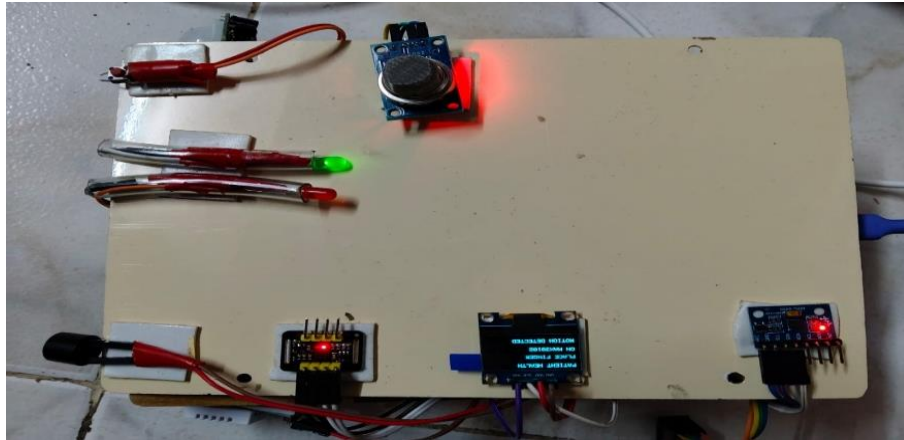
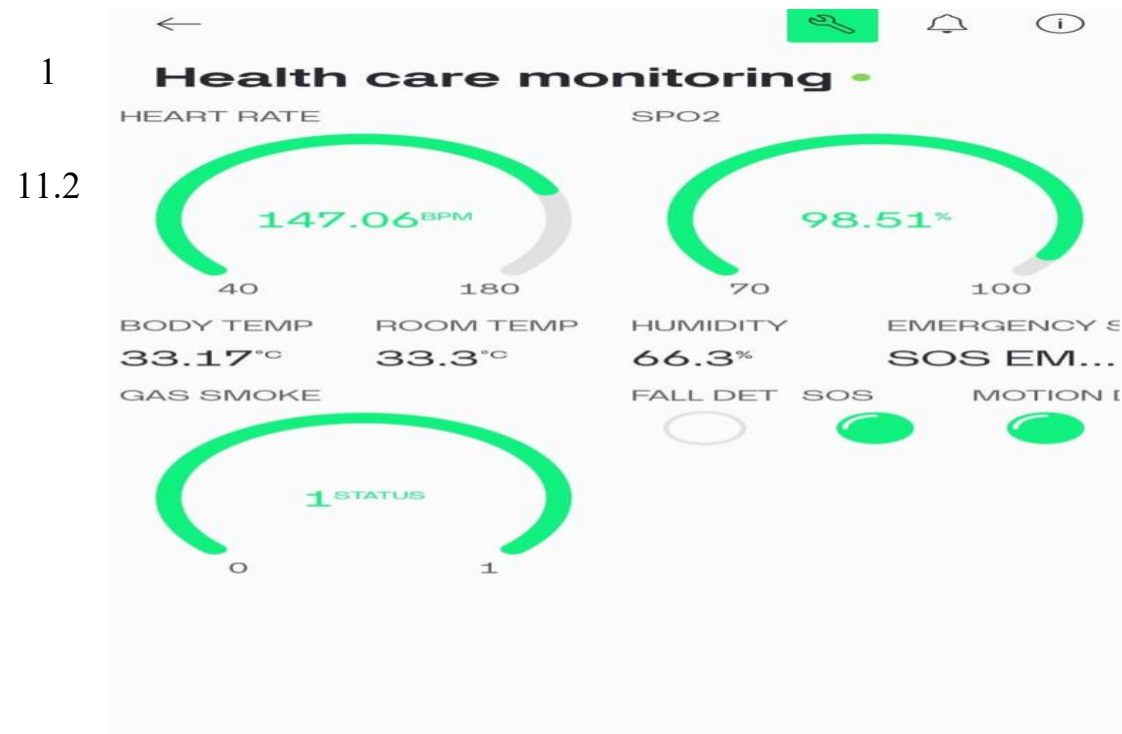


Figure 11.1

11.2 Blynk Dashboard



Figure

CHAPTER 12

CONCLUSION

The Intelligent Remote Healthcare and Assisted Living Ecosystem was developed to provide a simple and affordable solution for monitoring the health and safety of elderly people, patients, and individuals who may require regular assistance. The main objective of the project was to combine health monitoring, environmental monitoring, emergency alerts, remote access, and data storage into a single system.

The proposed system uses an ESP32 microcontroller to collect information from different sensors. The MAX30102 is used to monitor heart rate and SpO₂, while the MLX90614 measures body temperature. The MPU6050 helps detect sudden movements and possible falls. Environmental conditions are monitored using the DHT22, MQ-2, and PIR sensors, which provide information about room temperature, humidity, gas or smoke, and movement.

The collected readings are displayed on an OLED screen, allowing the user to view the information locally. At the same time, the data is transmitted through Wi-Fi to the Blynk IoT platform, making remote monitoring possible for family members or caregivers. The buzzer and LED indicators provide alerts when an abnormal condition is detected, while the SOS button gives the user a simple way to manually request assistance during an emergency.

The project also includes a Python-based data logging system that retrieves the monitored information from the Blynk IoT Cloud and stores it in a CSV file. This provides a useful record of previous readings and makes it possible to review and analyze the monitored data over time.

The testing of the prototype demonstrated that the major functions of the system could be successfully integrated and operated together. The sensors provided the required readings, the alert mechanism responded to simulated emergency conditions, Blynk enabled remote monitoring, and the Python program stored the collected data automatically.

Overall, the project demonstrates how IoT, embedded systems, and sensor technology can be combined to create a practical healthcare and assisted living solution. Although the prototype has limitations related to sensor accuracy, Wi-Fi connectivity, fall detection, and power supply, it provides a strong foundation for further development. With improvements such as better sensor calibration, advanced fall detection algorithms, battery operation, GPS, cellular communication, and mobile application support, the system can be developed into a more reliable and user-friendly solution for real-world applications.

In conclusion, the proposed system provides an affordable and flexible approach to remote healthcare and safety monitoring. It can help caregivers and family members stay informed about the user's condition and respond more quickly when an emergency or unusual situation occurs.

CHAPTER 13

REFERENCES

- [1] Sapci, A. Hasan, and H. Aylin Sapci. "Innovative assisted living tools, remote monitoring technologies, artificial intelligence-driven solutions, and robotic systems for aging societies: systematic review." *JMIR aging* 2, no. 2 (2019): e15429.
- [2] Ahmad, Ijaz, Zeeshan Asghar, Tanesh Kumar, Gaolei Li, Ahsan Manzoor, Konstantin Mikhaylov, Syed Attique Shah et al. "Emerging technologies for next generation remote health care and assisted living." *IEEE Access* 10 (2022): 56094-56132.
- [3] Qian, Kun, Zixing Zhang, Yoshiharu Yamamoto, and Björn W. Schuller. "Artificial intelligence internet of things for the elderly: From assisted living to health-care monitoring." *IEEE Signal Processing Magazine* 38, no. 4 (2021): 78-88.
- [4] Maskeliūnas, Rytis, Robertas Damaševičius, and Sagiv Segal. "A review of internet of things technologies for ambient assisted living environments." *Future Internet* 11, no. 12 (2019): 259.
- [5] Aalam, Javed, Syed Naseer Ahmad Shah, and Rafat Parveen. "Personalized healthcare services for assisted living in healthcare 5.0." *Ambient Assist. Living* (2025): 203-222.

- [6] Baldissera, Thais A., Cristiano De Faveri, Maria A. Oliveira, and Luis M. Camarinha-Matos. "Collaborative communication and monitoring ecosystem for elderly care." In Working Conference on Virtual Enterprises, pp. 34-49. Cham: Springer Nature Switzerland, 2024.
- [7] Uddin, Md Zia. "Smart assisted living: A data-centric and machine learning approach to intelligent home systems." IEEE Potentials (2026).
- [8] Sapci, A. Hasan, and H. Aylin Sapci. "Innovative assisted living tools, remote monitoring technologies, artificial intelligence-driven solutions, and robotic systems for aging societies: systematic review." JMIR aging 2, no. 2 (2019): e15429.
- [9] Abdi, Sarah, Luc de Witte, and Mark Hawley. "Emerging technologies with potential care and support applications for older people: review of gray literature." JMIR aging 3, no. 2 (2020): e17286.
- [10] Timon, Claire M., Emma Heffernan, Sophia Kilcullen, Louise Hopper, Hyowon Lee, Pamela Gallagher, Alan F. Smeaton, Kieran Moran, Pamela Hussey, and Catriona Murphy. "Developing independent living support for older adults using internet of things and AI-based systems: co-design study." JMIR aging 7 (2024): e54210.